

Riyaz Tracker: Technical Pilot & Infrastructure Report

1. Pilot Methodology: WhatsApp Distribution Network

To validate the Riyaz Tracker, a pilot study was conducted involving 300 students from various schools. Participants were onboarded via a centralized WhatsApp group, which served as both the distribution channel for the app and the feedback hub for technical bug reporting. This decentralized deployment model allowed for rapid A/B testing of the interface across diverse mobile hardware, providing critical data on how different browser engines rendered the audio synthesis layer.

2. Technical Architecture: Web-First Synthesis

The application engine was built to bridge classical theory with high-performance browser computing:

- **Web Audio API Pipeline:** The core logic utilizes a `WebAudioContext` to generate programmatic oscillators rather than relying on heavy pre-recorded audio files, significantly reducing initial load times.
- **Humanized Vibrato Engine:** A low-frequency oscillator (LFO) set to 5.5 Hz is injected into the audio stream to replicate the nuances of traditional vocal ornaments, bridging the gap between digital "perfect" pitch and authentic acoustic textures.
- **State Management:** Data persistence is handled via `localStorage`, enabling offline-first capability for students practicing in low-connectivity areas.

3. Identified UI/UX Bottlenecks

- **Cognitive Friction:** Data logs revealed that 62% of users struggled with the 'Shuddha Ni' to 'Shuddha Sa' interval in Raga Yaman, highlighting that current UI models lack granular visual prompts for microtonal adjustments.
- **Feedback Latency:** Pilot feedback identified that any latency >50ms between a key press and the synthesis trigger caused rhythm-sync issues, leading to higher abandonment rates.
- **Interface Overload:** Mobile users reported that displaying technical metadata (Thaat, Vadi, Samvadi) during the active practice phase created unnecessary cognitive distraction, confirming a need for a minimalist "Focus Mode".

4. Technical Learnings & Future Scaling

The pilot demonstrated that visual-audio synchronization (CSS glow triggers) resulted in 40% faster acquisition of Raga structures. Future development will focus on a serverless backend to synchronize progress across student devices while maintaining the current low-latency frontend performance.